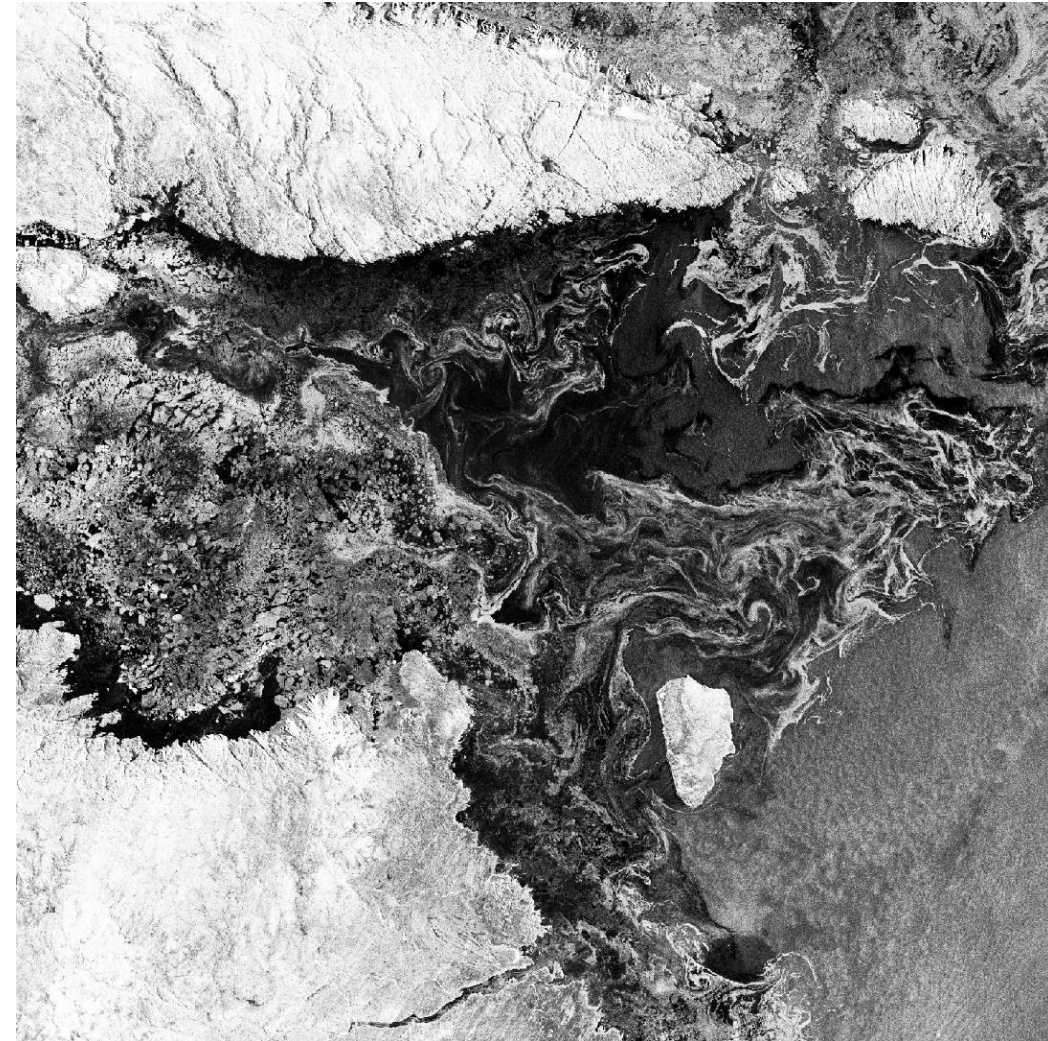
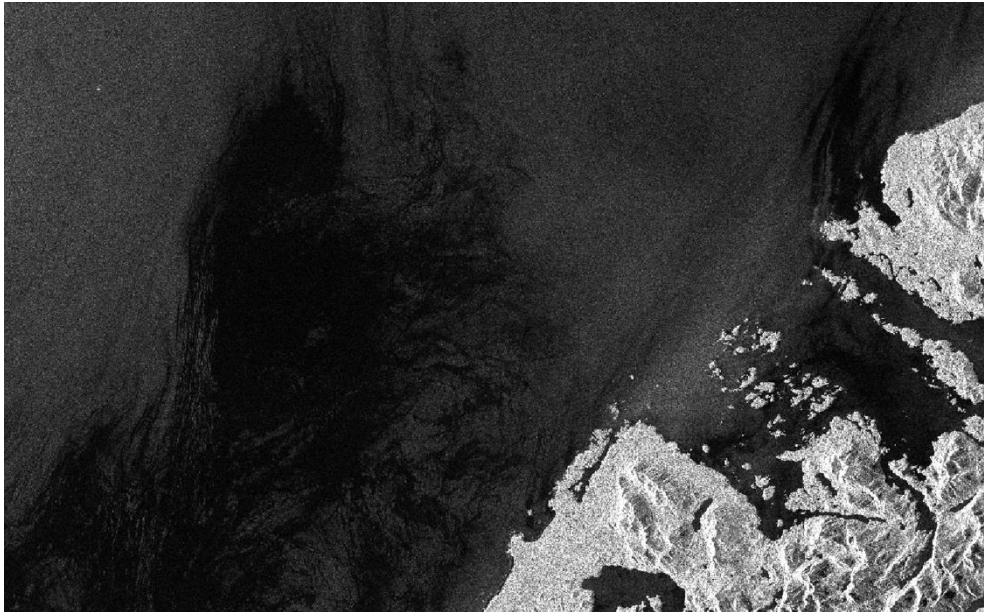


Synthetic aperture radar

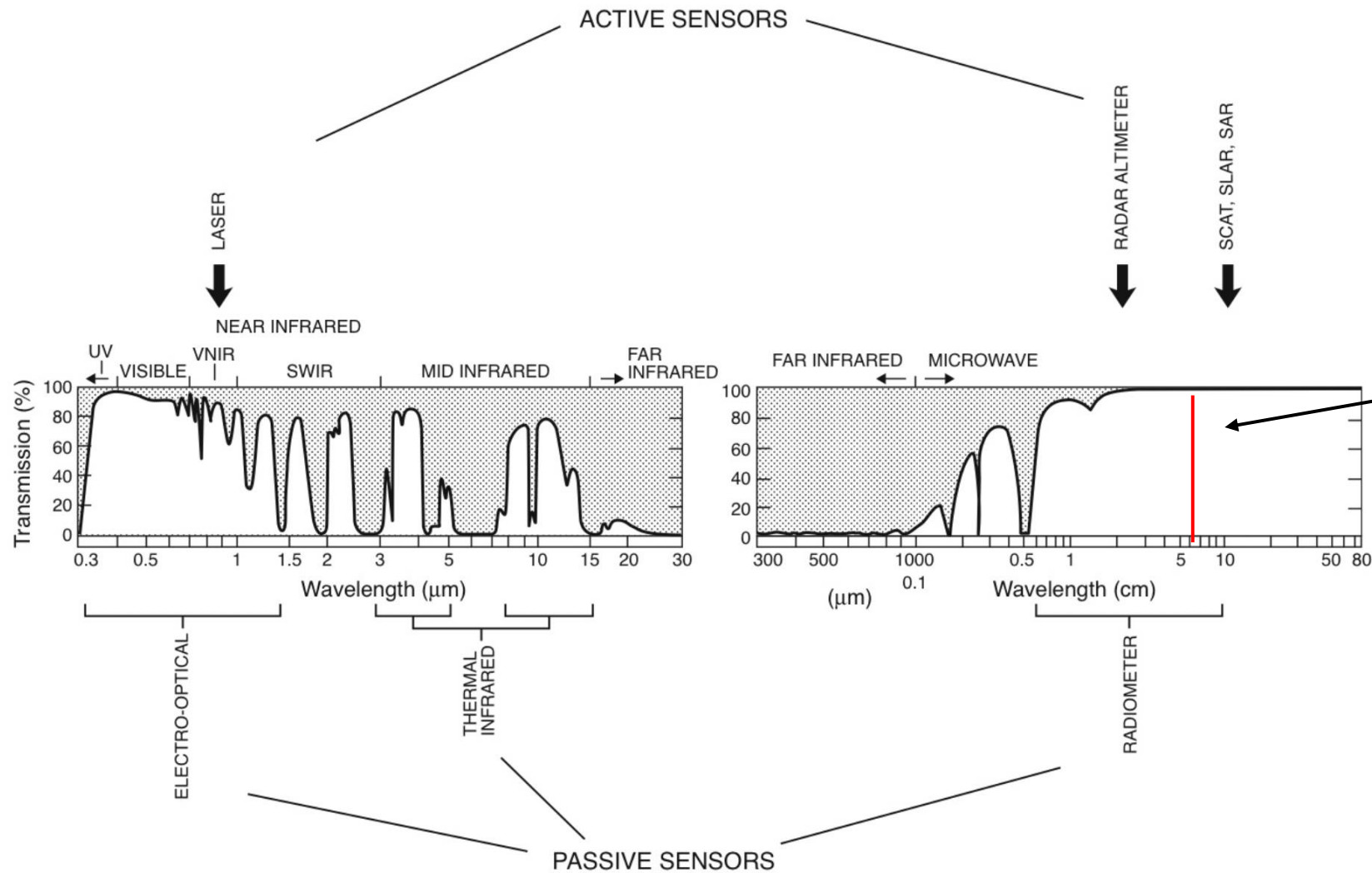
Introduction to the fundamentals

What is SAR?

- High spatial resolution, low frequency microwave
- an **active** sensing system
- sends out a pulse of low frequency microwaves, and records the returning energy
- Rich textures, multiscale information

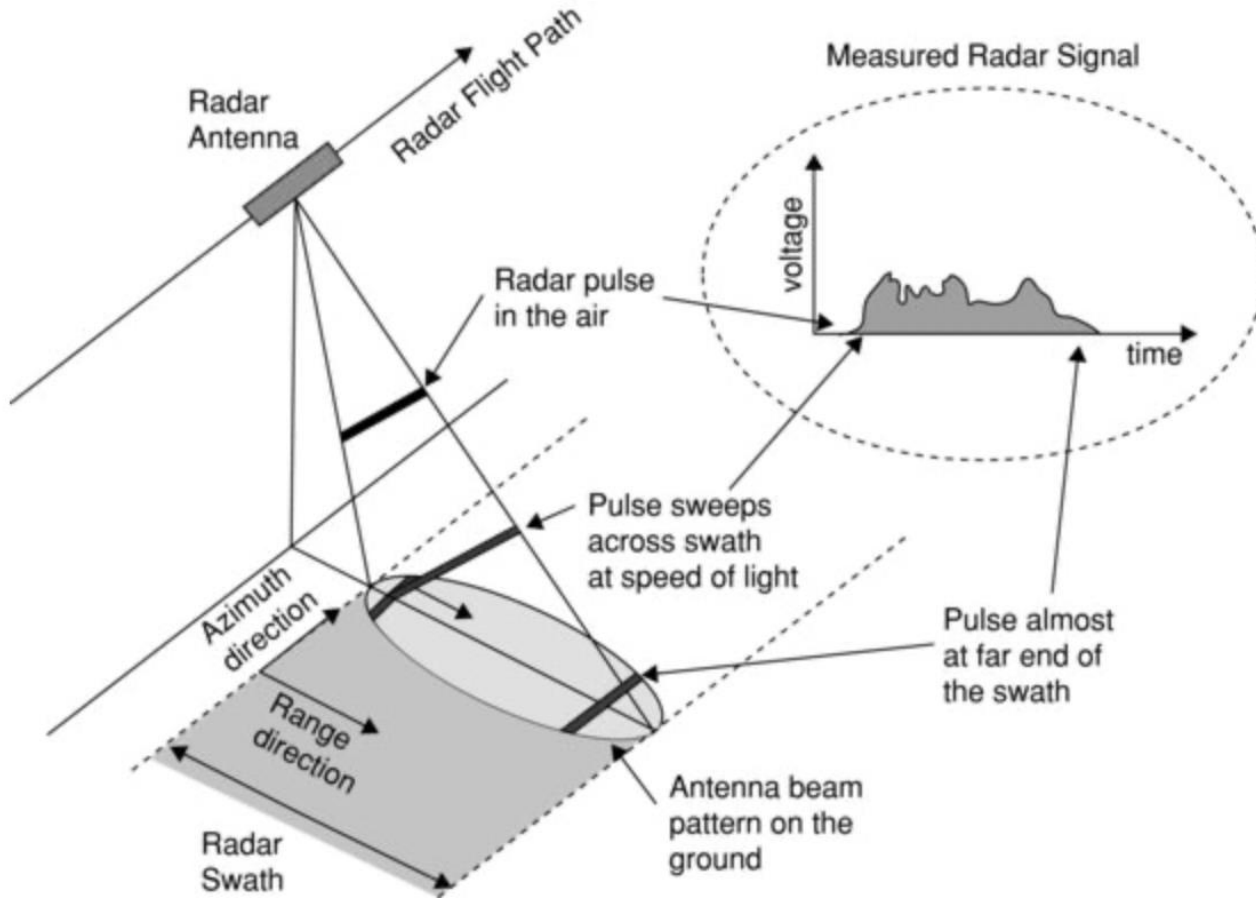


Recall the electromagnetic spectrum:



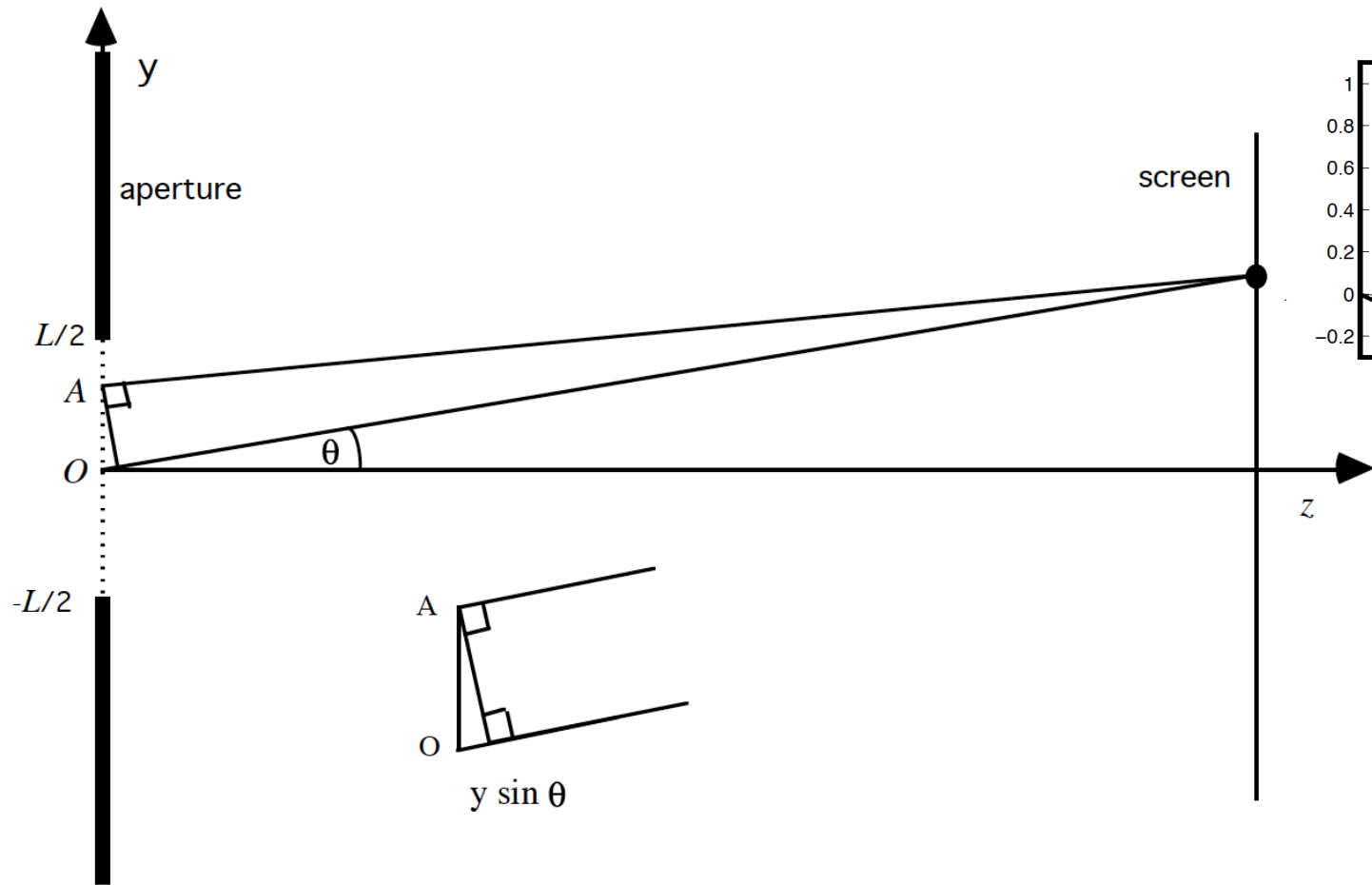
C-band SAR,
most satellite-based SAR
sensors are in the 1-40 GHz
range or 30 cm to 8 mm
wavelengths

Imaging geometry

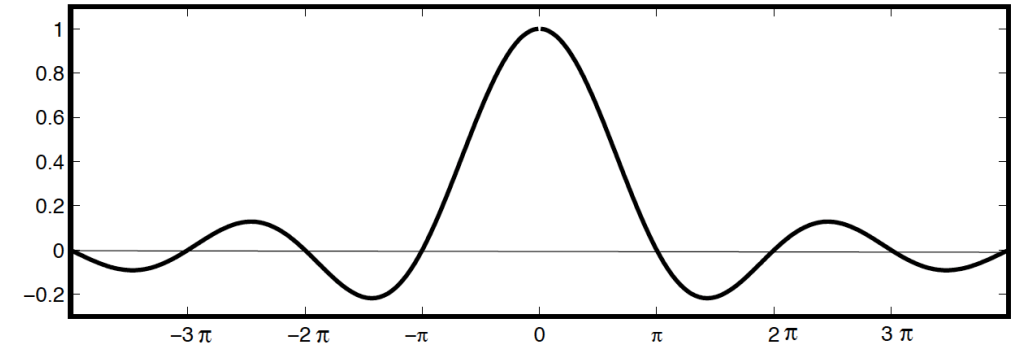


- Here shown for side looking SAR (e.g. Sentinel-1, Radarsat sensors, SWOT is different)
- Azimuth – along-track
- Range – across track

For the imaging pattern: recall diffraction

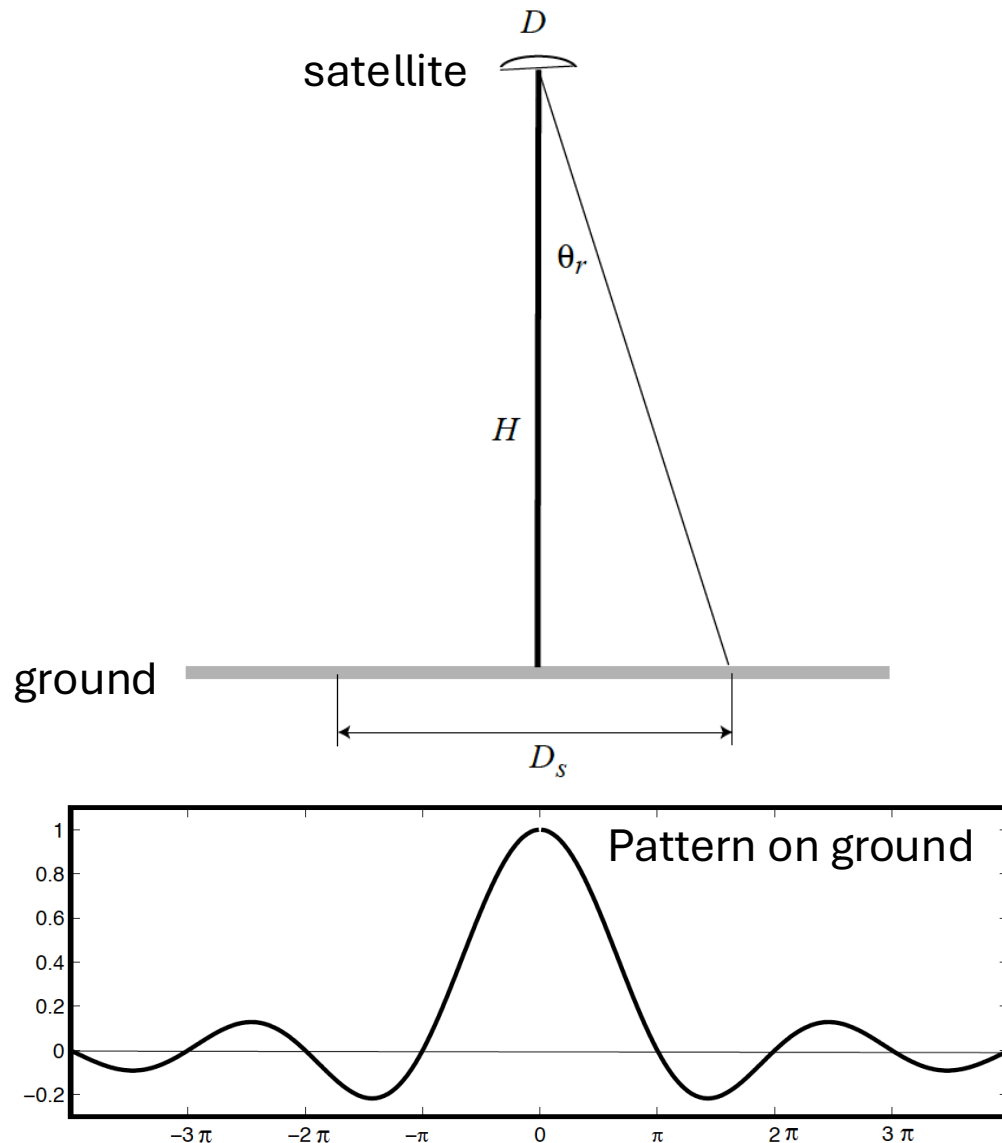


Pattern on screen



For a satellite sensor, L is the antenna size

How does this apply to the satellite ?



$$P(\theta) = L \operatorname{sinc}\left(\frac{L\pi \sin \theta}{\lambda}\right)$$

$$\text{Zero crossing: } \sin(\theta) = \frac{\lambda}{L}$$

$$\text{Small angle: } \sin(\theta) \approx \tan(\theta)$$

$$\sin(\theta) = \frac{Ds}{2H} = \frac{\lambda}{L} \quad Ds = 2H\lambda/L$$

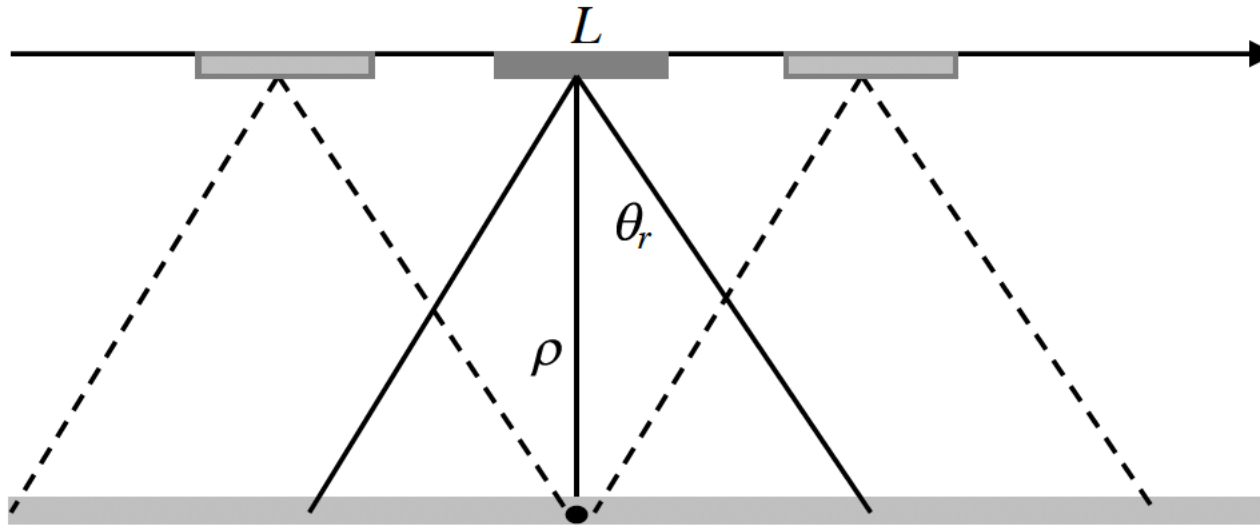
Let our SAR instrument have $\lambda = 5\text{cm}$

Let the altitude $H = 800\text{ km}$

If the antenna is 1m in length, $Ds = 40\text{ km}$

But we want Ds to be closer to 40 m
Limitation is L – antenna size

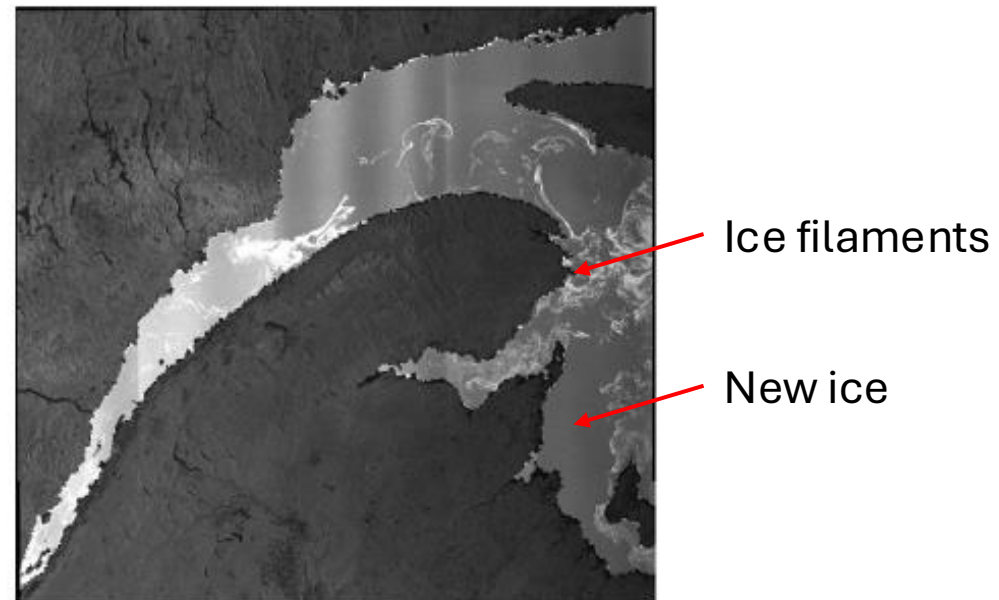
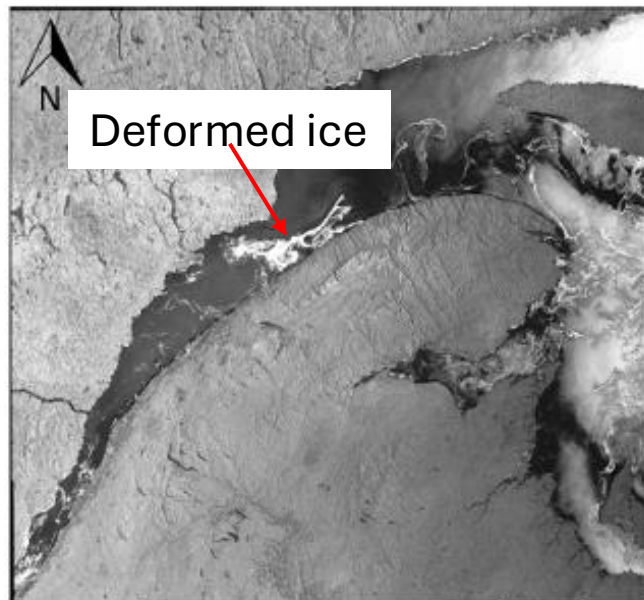
How does SAR achieve high spatial resolution?



A larger antenna is **synthesized** using the movement of the spacecraft

How are SAR data different?

- They send **coherent** waves -> waves have the same frequency and constant phase shift (similar to lasers)
- They receive the reflected waves from the ground or target
- They can have rich polarization information about the target

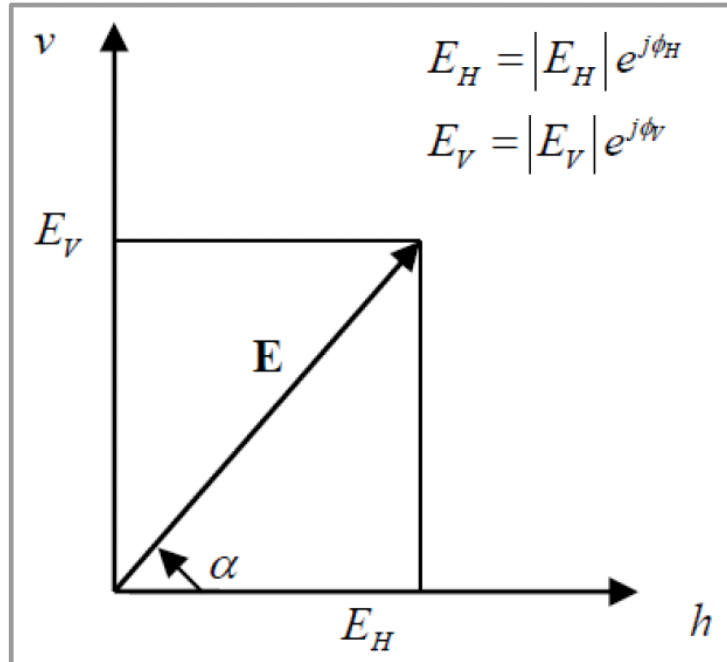


HH: Horizontal send – horizontal receive

HV: Horizontal send – vertical receive

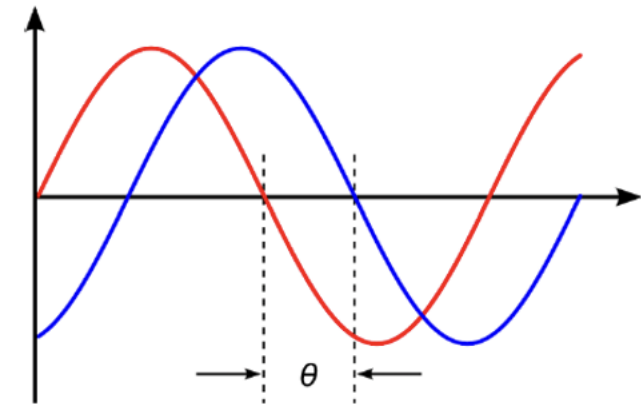
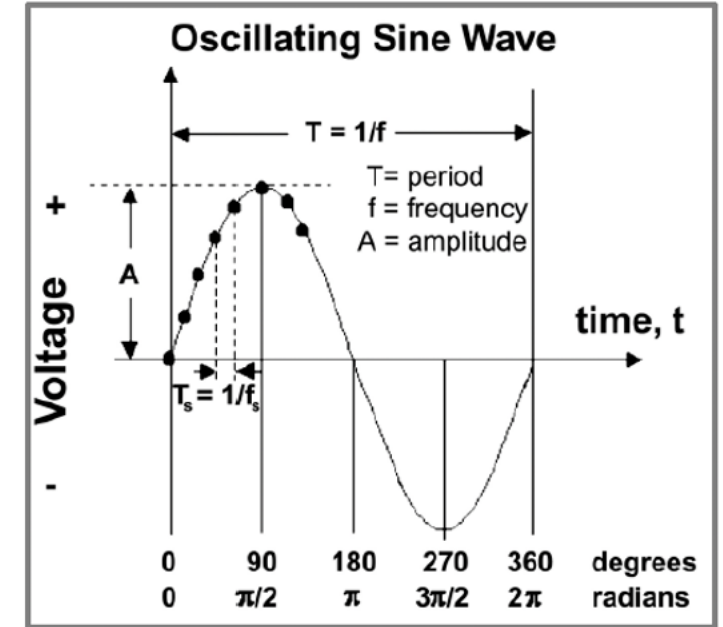
How are SAR data represented?

- Recall some basics of a wave
- Phase:** the position of a point in the wave cycle
- Amplitude:** maximum displacement from rest
- They are represented as complex numbers



$|E|$ = amplitude
 ϕ = phase

A **complex number** is a number that can be expressed in the form $a + bi$, where a and b are real numbers, and i is a solution of the equation $x^2 = -1$.



A consequence of coherent imaging

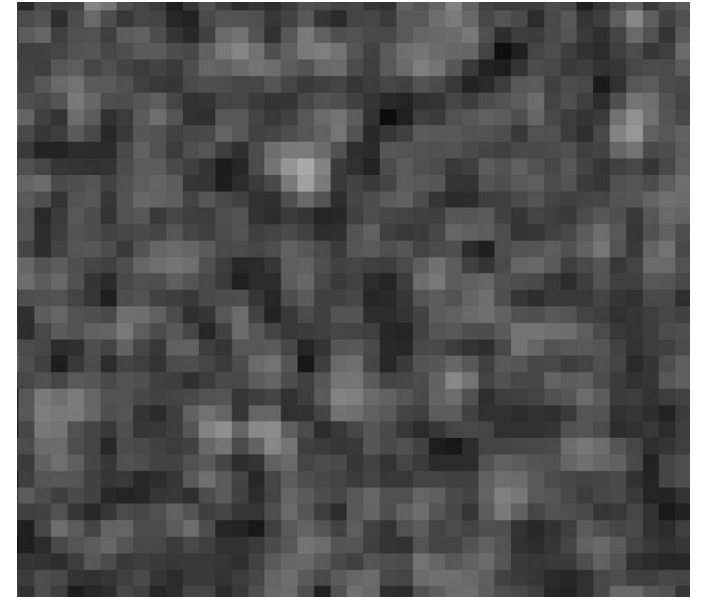
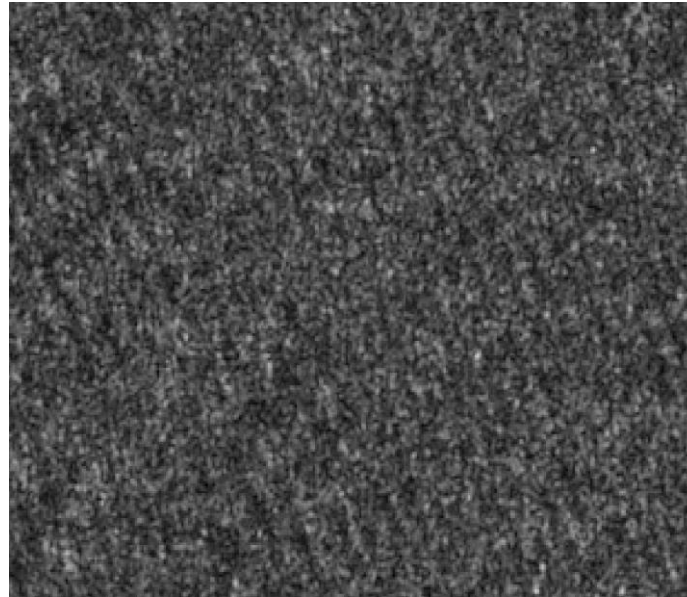
- Coherent imaging leads to speckle noise/salt and pepper noise
- Within a measurement pixel there are multiple small scatterers
- The resulting signal is a sum of these

$$S = \sum_i A_i e^{j\phi_i}$$

- The system will measure both the amplitude and phase
- Speckle results due to constructive and destructive interference patterns

Seeing speckle

Increasing zoom



On a calm day over open water, the waves and swell dominates. Waves that are on the order of the SAR wavelength (approx. 5 cm) will contribute significantly to the radar return (Bragg scattering)

A consequence of side looking geometry

- Intensity can vary significantly across the swath
- Sometimes incidence angle normalization is done
- Normalization will be different for different surface types (water, ice etc.)

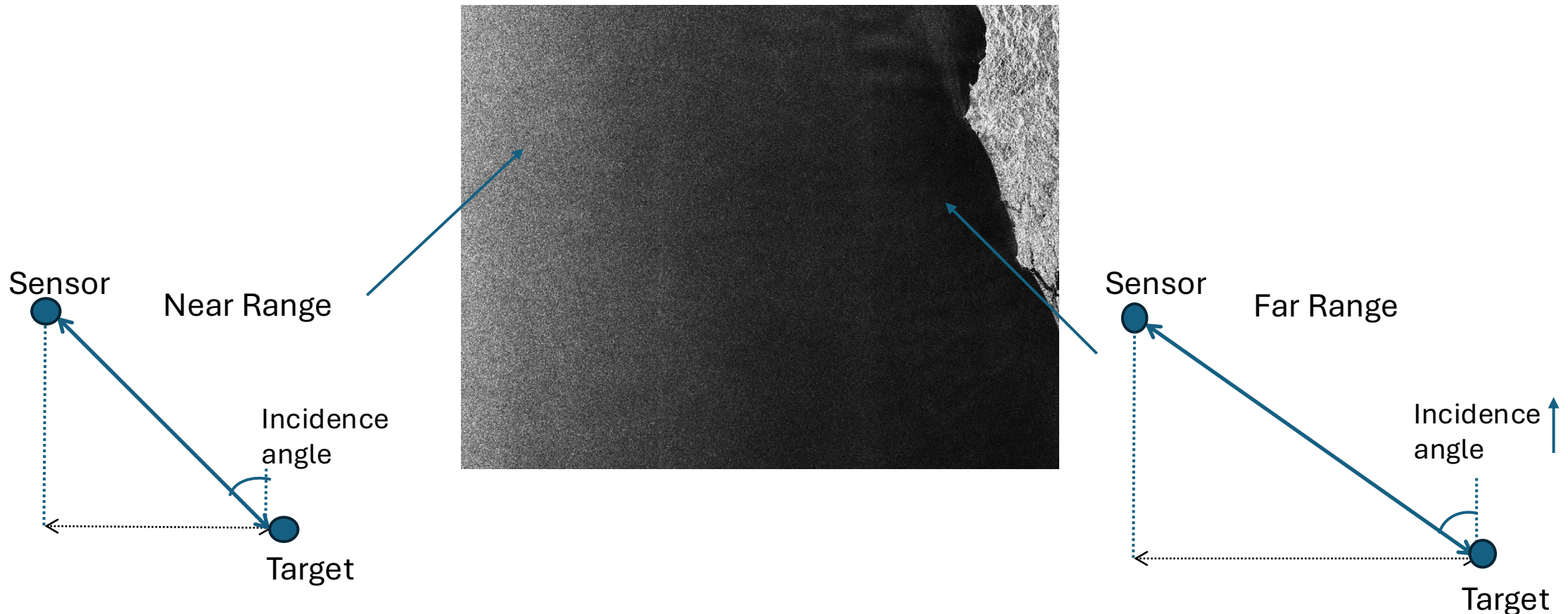
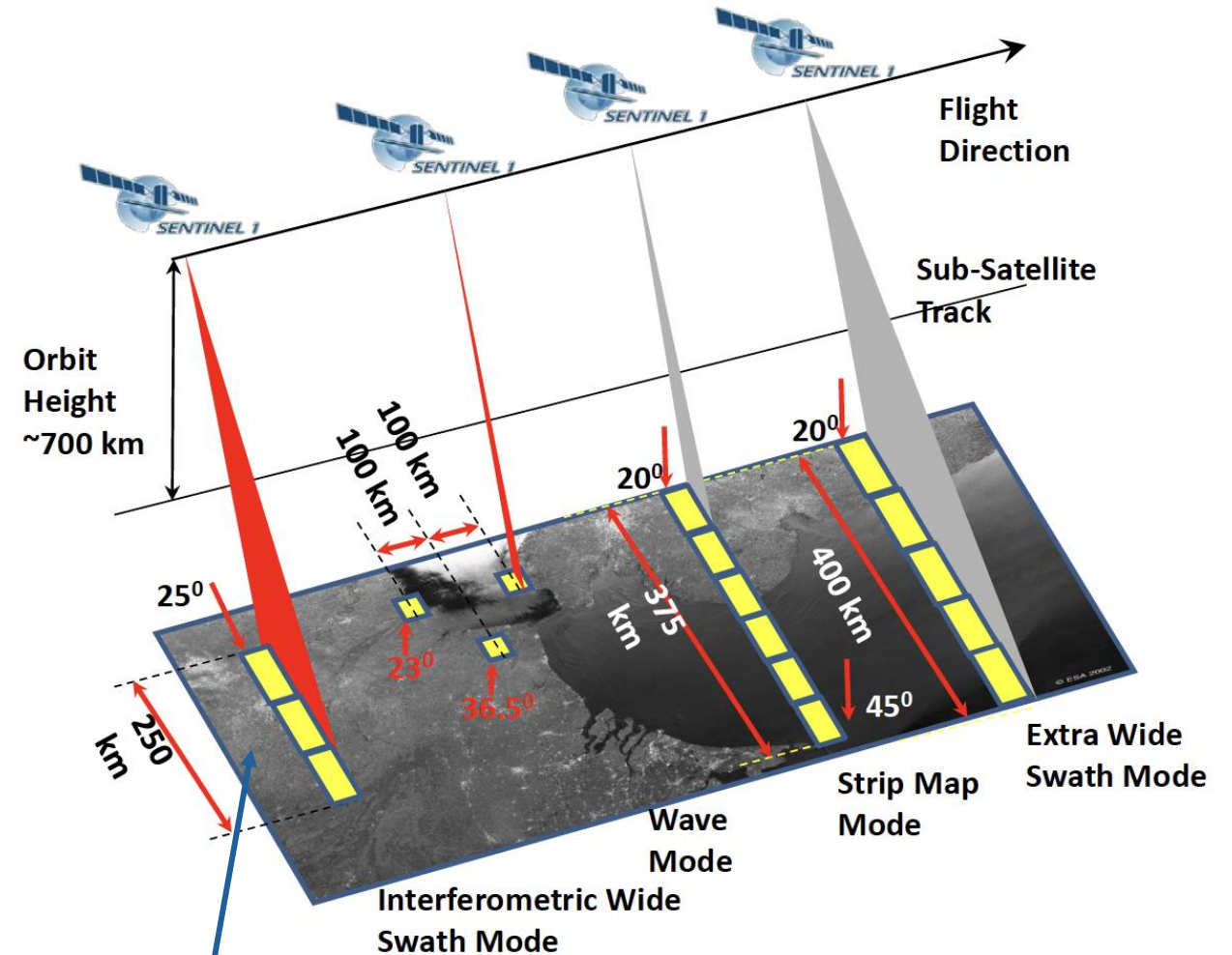
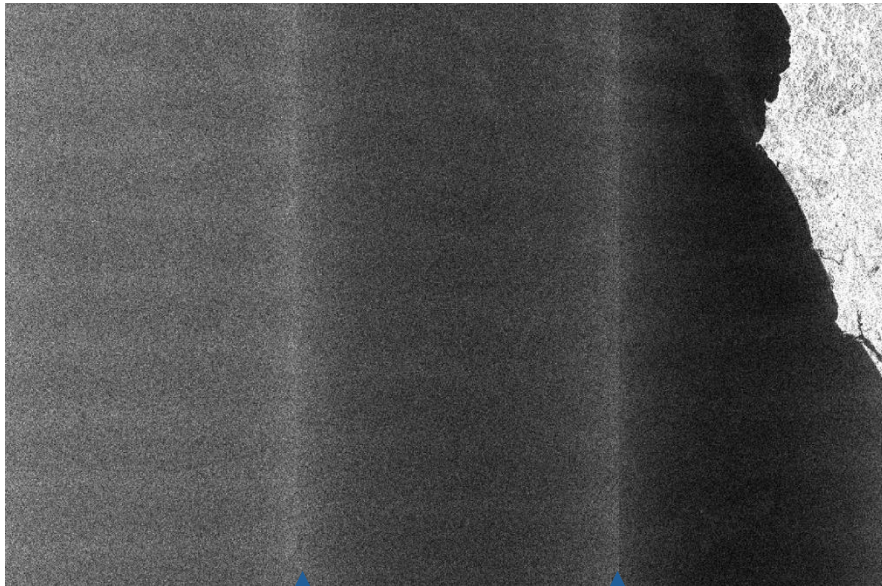


Image artifacts (processor level)

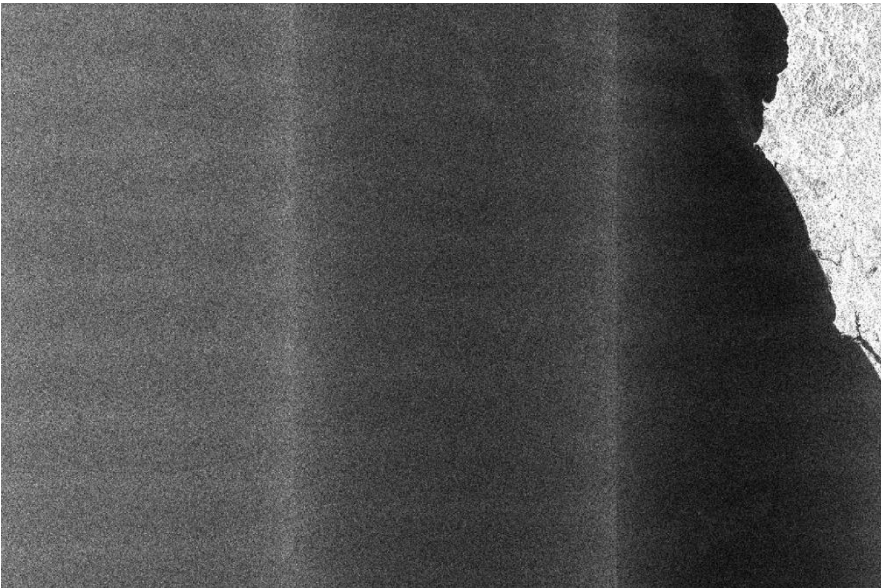
- Tasked sensor: Different modes scheduled for different purposes
- Swath width and spatial resolution vary by mode



Sources: ESA SAR Tutorial

Image artifacts (processor level)

- When subswaths are stitched together there are banding artifacts that can be very noticeable



- This image is over the ocean
- There is strong intensity variation with incidence angle
- In part it is brighter in the near range because the incidence angle is lower so the radar is hitting the surface more directly.

Calibration

- Radar backscatter is the ratio of scattered power to incident power
- The backscatter will be different depending on the area that is sensed
- The area depends on the sensor and the imaging geometry
- Generally one can calibrate using different reference areas

Slant reference area

$$A_\beta, \rightarrow \beta^o$$

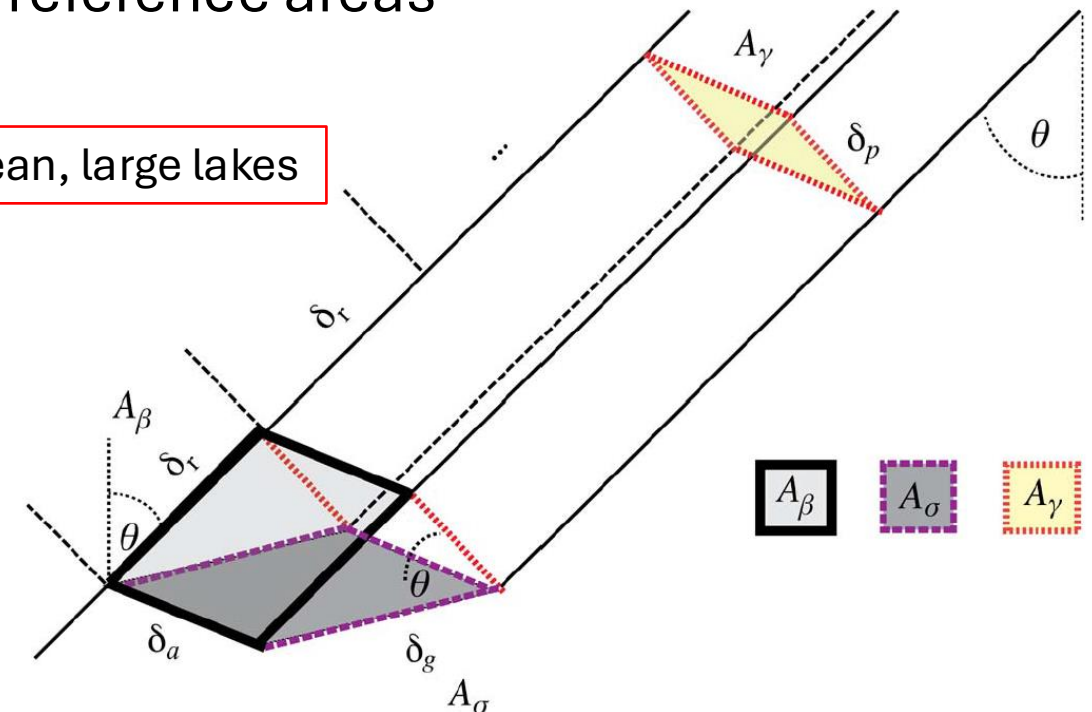
Used for ocean, large lakes

Ground reference area

$$A_\sigma, \rightarrow \sigma^o$$

Perpendicular reference area

$$A_\gamma, \rightarrow \gamma^o$$



SAR by example

- There is a great deal of theory behind SAR imaging
- Here we will cover some basics, and show some examples
- We will use SNAP – the Sentinel processor (free)

<https://step.esa.int/main/download/snap-download/>

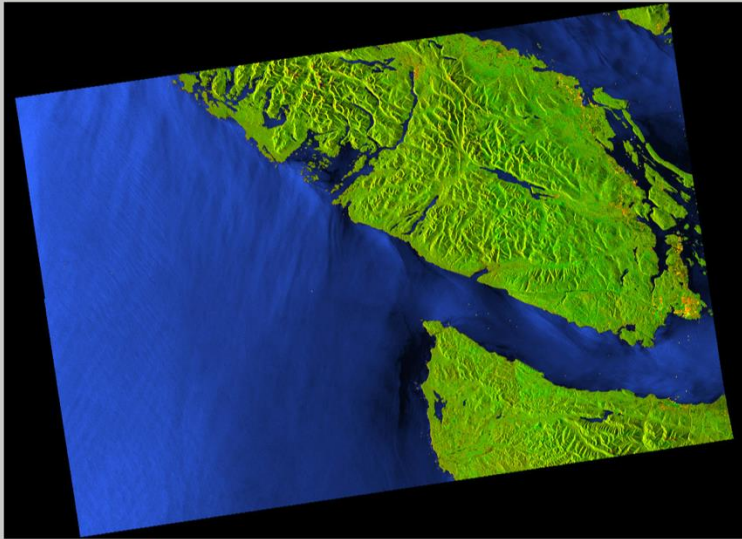
Data download: Alaska SAR Facility

The screenshot displays the ASF Data Search Vertex web application. At the top, the search criteria are defined: Search Type is 'Geographic Search', Dataset is 'Sentinel-1', and Area of Interest is a polygon in the Pacific Northwest. The Start Date is 6/4/2025 and the End Date is 7/26/2025. A red circle highlights these date fields, with a red arrow pointing to the word 'Date' in red text. Another red circle highlights a specific geographic area on the map (around Vancouver, BC), with a red arrow pointing to the word 'Region' in red text. The map view shows a satellite image of the Pacific Northwest with a blue polygon representing the search area. Below the map, there is a list of search results. The first result is 'S1C_IW_GRDH_1SDV_20250606T020946_20250606T021011_002656_0057_DE_8A9A' from Sentinel-1, C-Band, dated June 6, 2025. The interface also shows a 'Scene Detail' section for the selected scene, including metadata like Start Time, Stop Time, Flight Direction, and Polarization. On the right, there is a list of data products available for the selected scene, such as 'L1 Single Look Complex (SLC)', 'L0 Raw Data (RAW)', 'L1 Detected High-Res Dual-Pol (GRD-HD)', and 'L2 Ocean (OCN)'. The interface is clean and professional, with a clear layout for searching and viewing satellite data.

Visualizing the image acquisitions

S1C_IW_GRDH_1SDV_20250606T020946_20250606T021011_002656_0057DE_8A9A

Sentinel-1 • June 6, 2025, 02:09:46Z



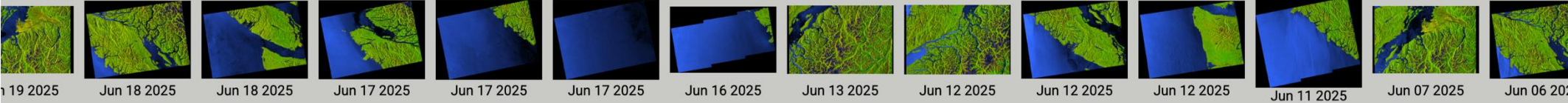
Scene Detail

Start Time • 06/06/2025, 02:09:46Z ⚙
Stop Time • 06/06/2025, 02:10:11Z ⚙
Beam Mode • IW ⚙
Path • 35 ⚙

Frame • 155 ⚙
Flight Direction • ASCENDING ⚙
Polarization • VV+VH ⚙
Absolute Orbit • 2656
PGE Version • 003.91
[Data courtesy of ESA](#)
[Citation](#)

8 Files

- L1 Single Look Complex (SLC)
- L0 Raw Data (RAW)
- L1 Detected High-Res Dual-Pol (GRD-HD)
- L2 Ocean (OCN)
- XML Metadata (SLC)
- XML Metadata (GRD-HD)
- XML Metadata (OCN)
- XML Metadata (RAW)



Only display scenes with a browse image

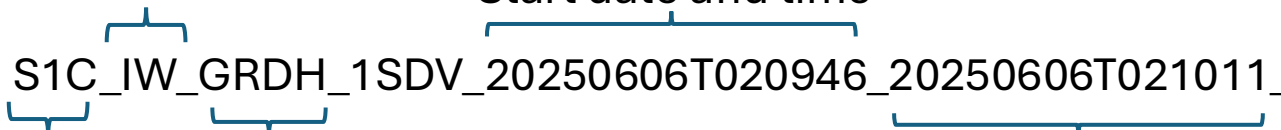
Close

Data downloaded:

Interferometric
wide mode

S1C_IW_GRDH_1SDV_20250606T020946_20250606T021011_002656_0057DE_8A9A

Sentinel 1C Mission Ground range detected High resolution End date and time



<https://sentiwiki.copernicus.eu/web/s1-products>

Sentinel 1: Choosing your data

[https://sentiwiki.copernicus.eu/
web/s1-products](https://sentiwiki.copernicus.eu/web/s1-products)

Acq. Mode	Product Type	Resolution Class	Resolution Rng x Azi [m]	Pixel Spacing Rng x Azi [m]	Num Looks Rng x Azi	ENL
SM	SLC	-	1.7x4.3 to 3.6x4.9	1.5x3.6 to 3.1x4.1	1x1	1
	GRD	FR	9x9	3.5x3.5	2x2	3.7
		HR	23x23	10x10	6x6	29.7
		MR	84x84	40x40	22x22	398.4
IW	SLC	-	2.7x22 to 3.5x22	2.3x14.1	1x1	1
	GRD	HR	20x22	10x10	5x1	4.4
		MR	88x87	40x40	22x5	81.8
EW	SLC	-	7.9x43 to 15x43	5.9x19.9	1x1	1
	GRD	HR	50x50	25x25	3x1	2.8
		MR	93x87	40x40	6x2	10.7
WV	SLC	-	2.0x4.8 3.1x4.8	1.7x4.1 2.7x4.1	1x1	1
	GRD	MR	52x51	25x25	13x13	123.7

Data download

Product Explorer x Pixel Info x [1] Amplitude_VV x [1] Intensity_VH x [1] Amplitude_VH x [1] Intensity_VV x

[1] S1C_IW_GRDH_1SDV_20250606T020946_20250606T0210

- > Metadata
- > Vector Data
- > Tie-Point Grids
- > Quicklooks
- > Bands
 - Amplitude_VH
 - Intensity_VH
 - Amplitude_VV
 - Intensity_VV

Polarizations
VV, VH

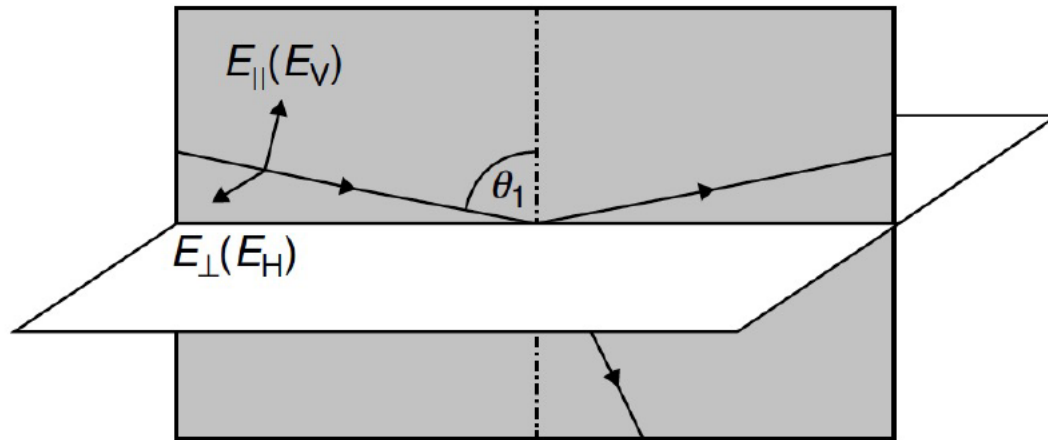
Amplitude:
Signal magnitude

Intensity:
Amplitude squared
(power)

Upside down?
No, Ascending pass

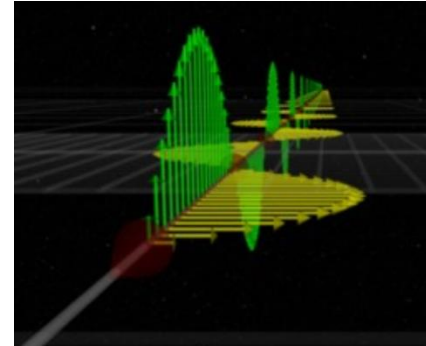
Tonal gradient

Linear Polarization



Rees, 2012

<https://svs.gsfc.nasa.gov/4580>



(this is actually showing the electric and magnetic waves, but the way they are orthogonal is similar for H and V, you could think of the green one as being in plane, or vertically polarized)

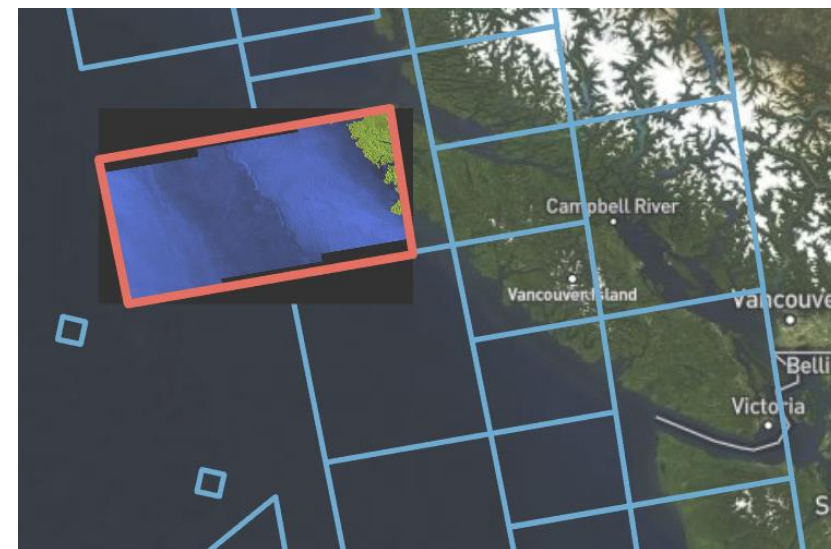
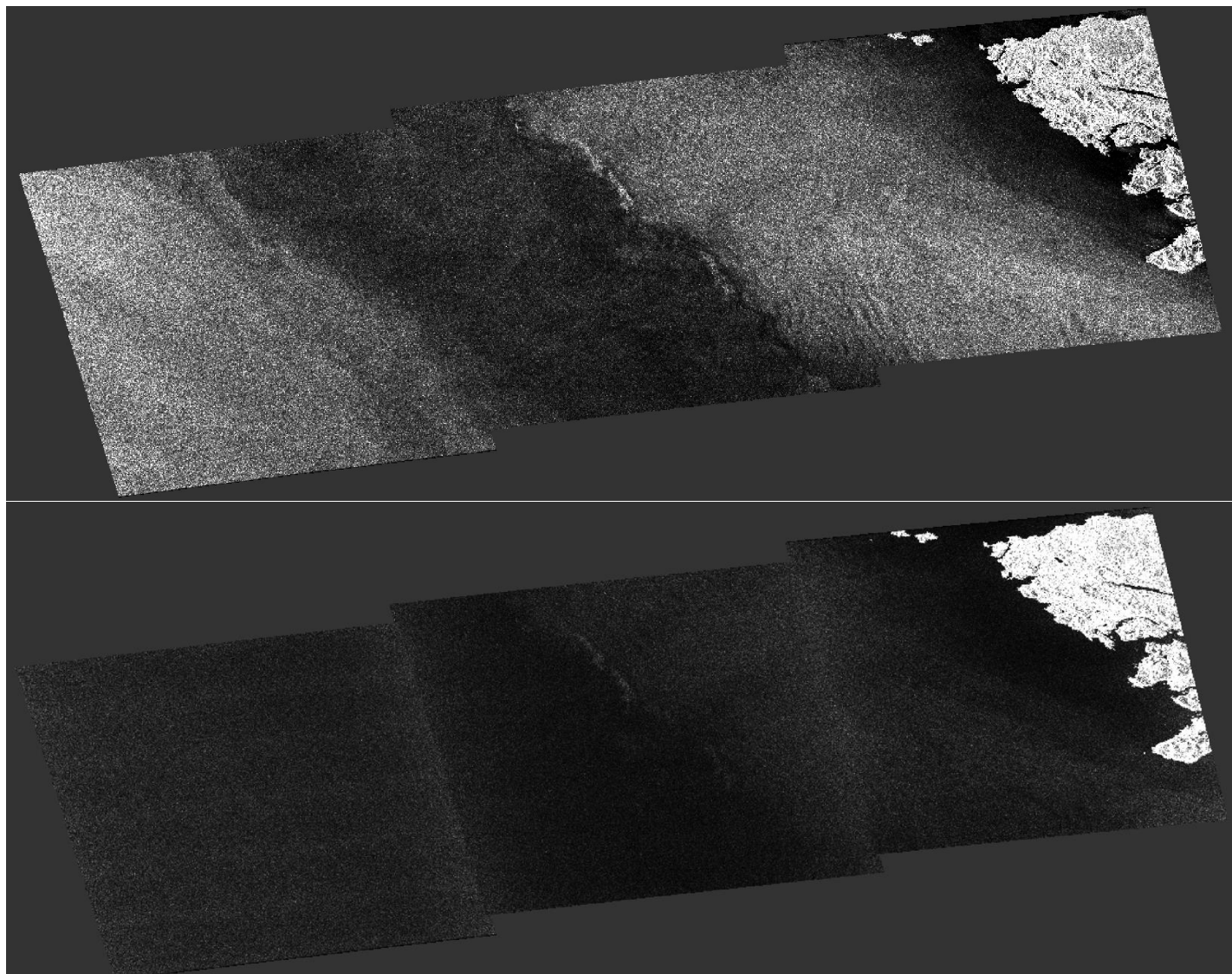
- Vertical polarization (E_V) Vpol
- Horizontal polarization (E_H) Hpol
- Both Vpol and Hpol components lie in the same plane, which is normal to the direction of travel

Polarization and SAR data

- SAR sensors send and receive polarized waves
- Dual pol are the most common
- Fully polarimetric data have more information
- There is also circular polarization (RADARSAT constellation mission RCM)

		Receive Polarization	
Transmit Polarization		H or V	H and V and relative phase
	H or V	Single [1] Pol	Dual [2] Pol Dual Polarimetric
	H and V	Dual [2] pol	Quad [4] Pol Fully Polarimetric

Dual pol Example: May 11, 2025



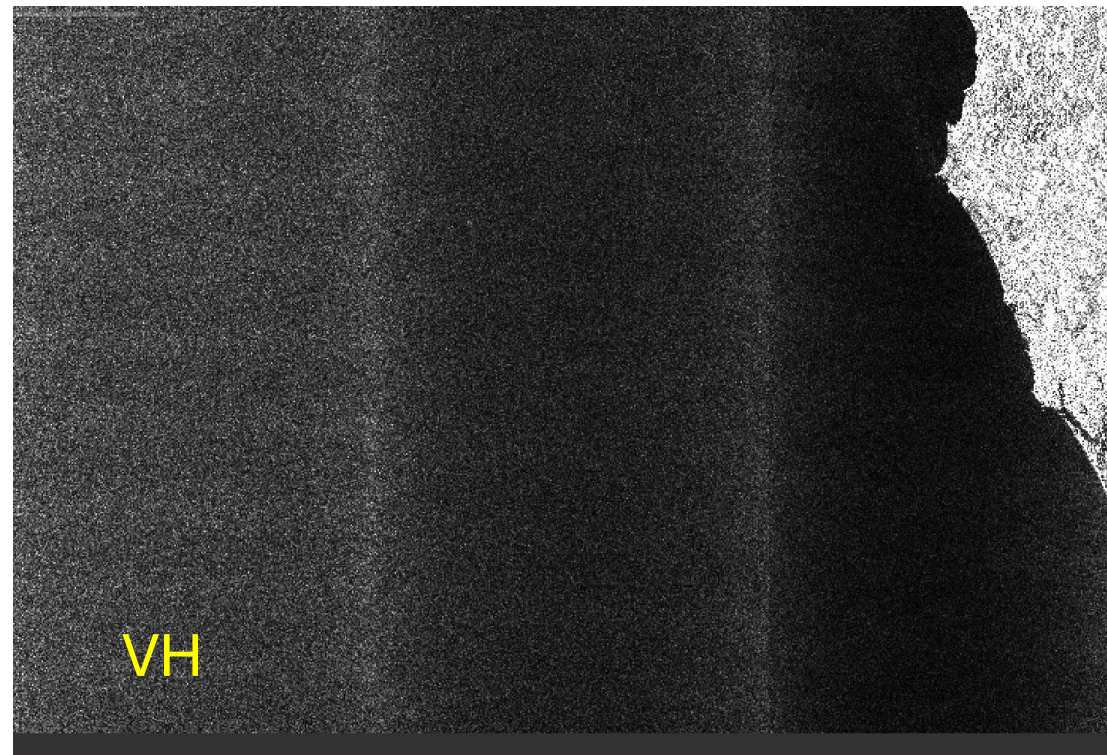
- VH is usually darker than VV and backscatter is closer to the noise floor
- Image bands are more prominent

Processing the data

snap File Edit View Analysis Layer Vector Raster Optical **Radar** Tools Window Help

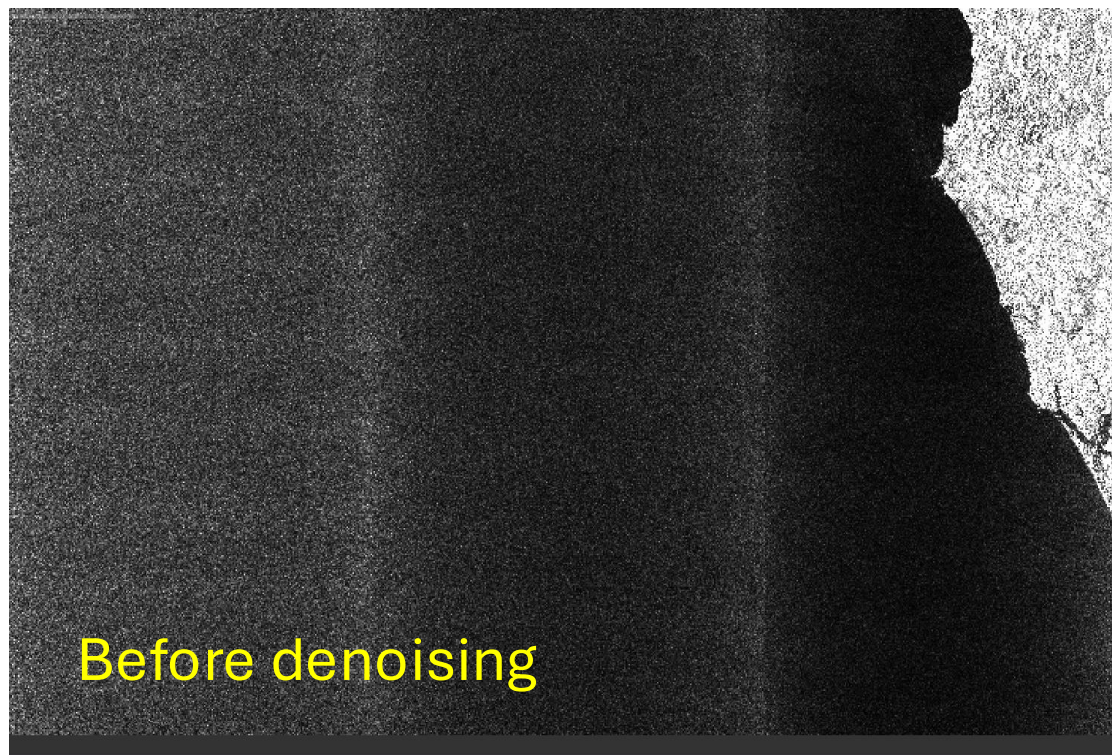
Step	Menu	Reason
Orbit correction	Radar -> Orbit correction	Satellite orbit is perturbed due to drag, solar wind, other objects, and needs a small correction
Thermal denoising	Radar -> Radiometric -> S1 Thermal Noise Removal	Banding in the images can lead to issues with interpretation
Calibration	Radar -> Radiometric -> Calibrate	Converts measurements into physically meaningful values, for the ocean you can convert to ground range (sigma)
Geometric Correction	Radar -> Radiometric -> Ellipsoid correction	Accurate geolocation, needed for comparison with other data sources and mapping, orientates image with latitude and longitude

Processing Example: 2025-04-18



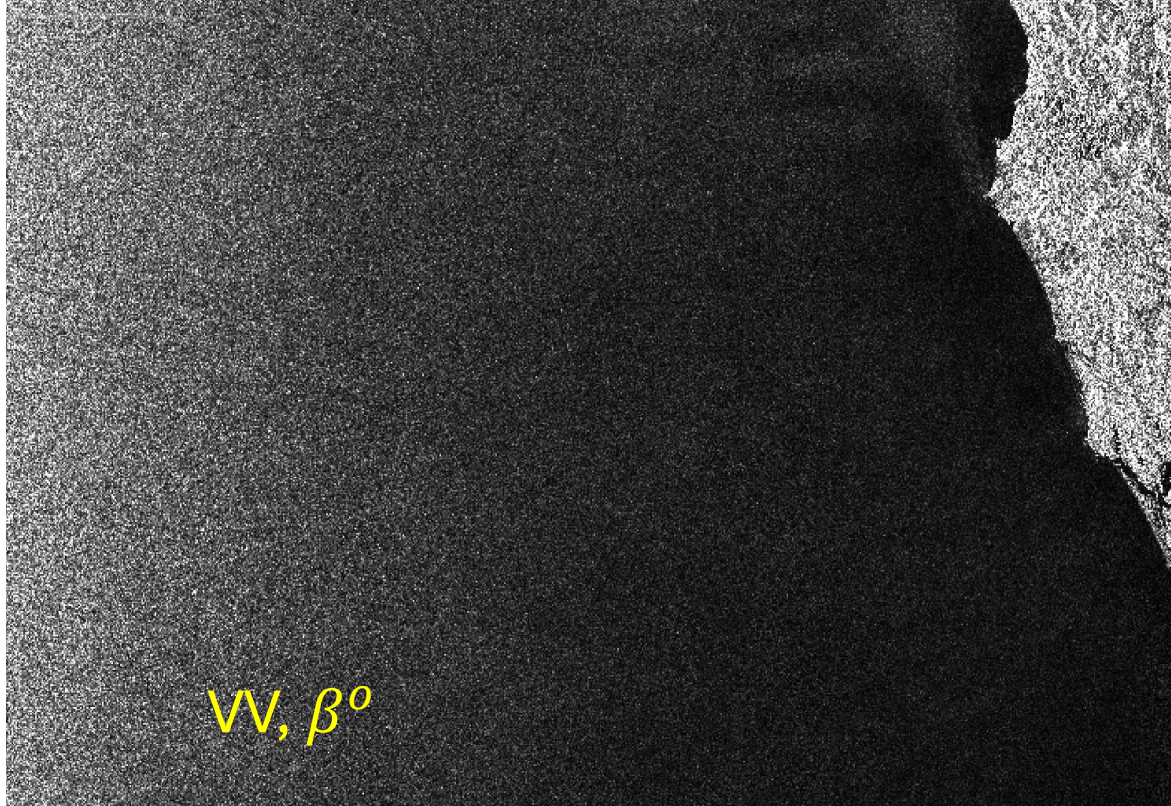
- This is the data downloaded Write down some of your observations

Thermal noise removal



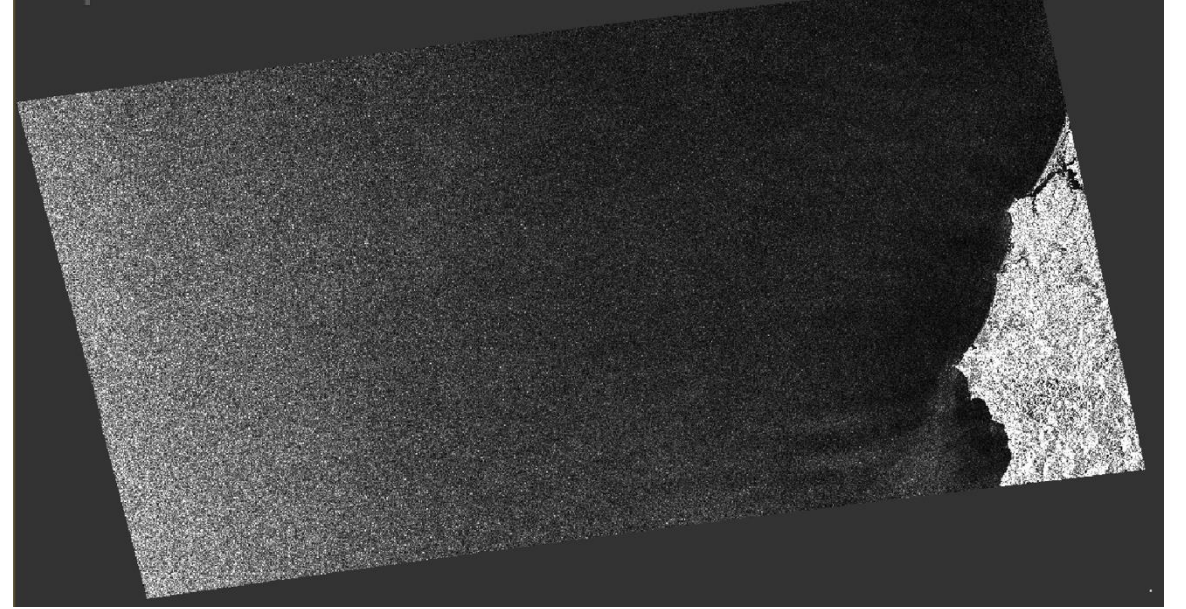
- Write down some of your observations

Calibration

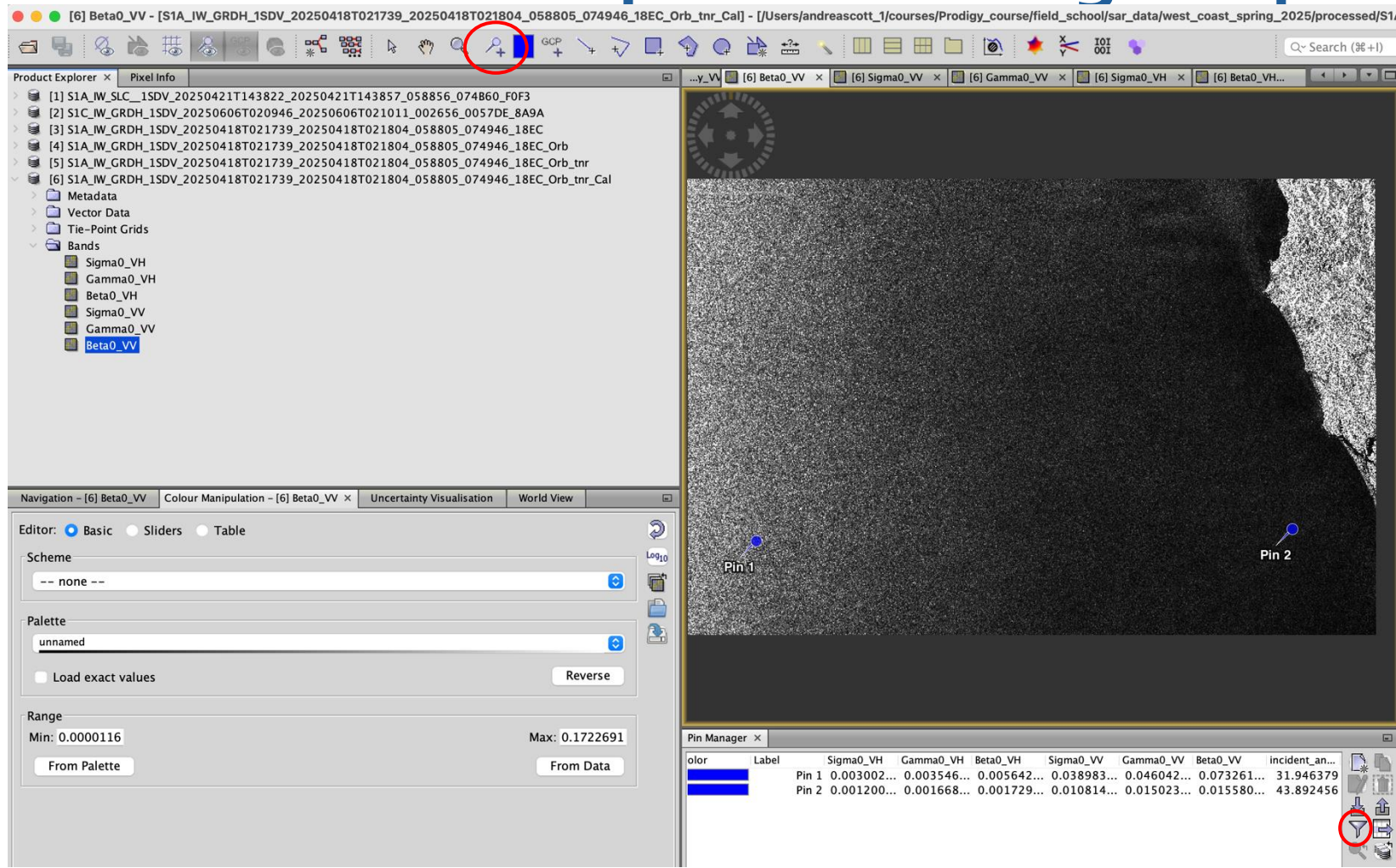


- Write down your observations

Geometric correction

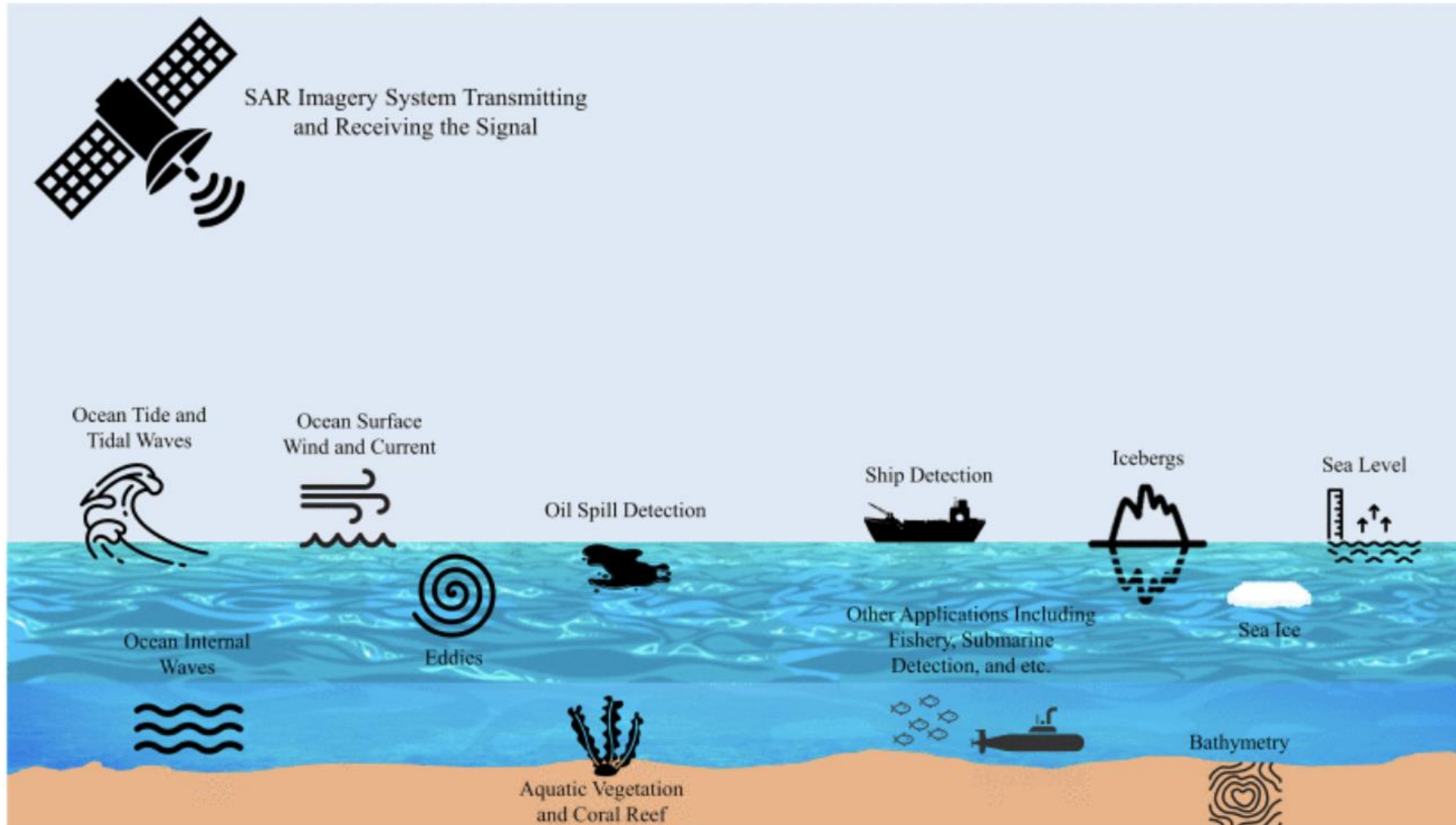


Quantitative comparison: Using the pin tool:



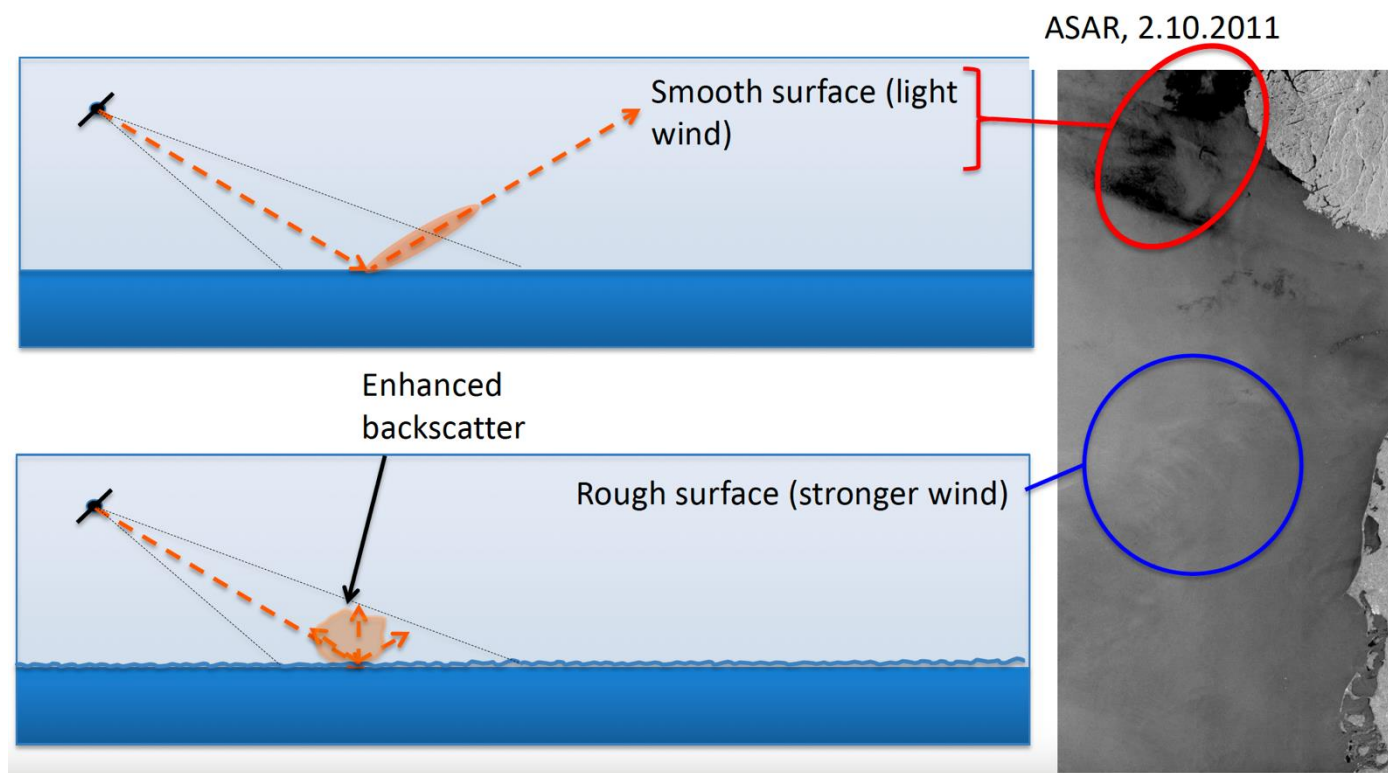
Filter to get values

What can we look at with SAR? (for the ocean...)



Interpreting SAR imagery

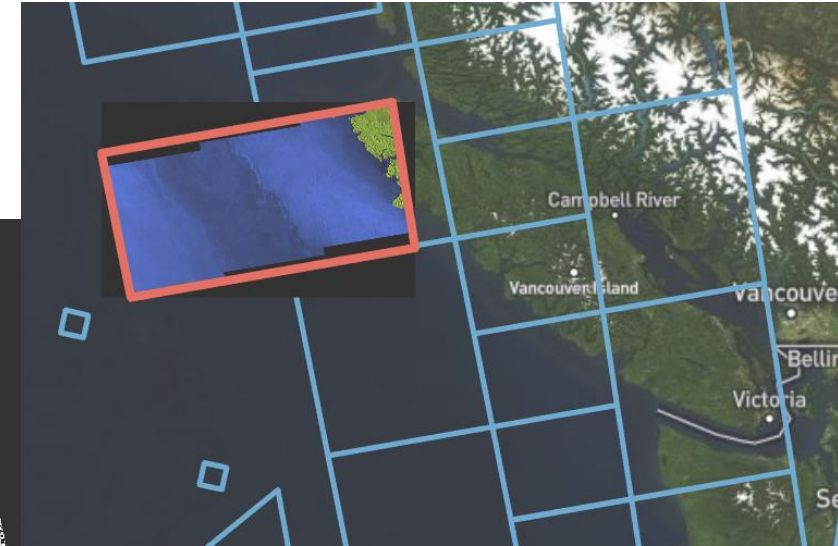
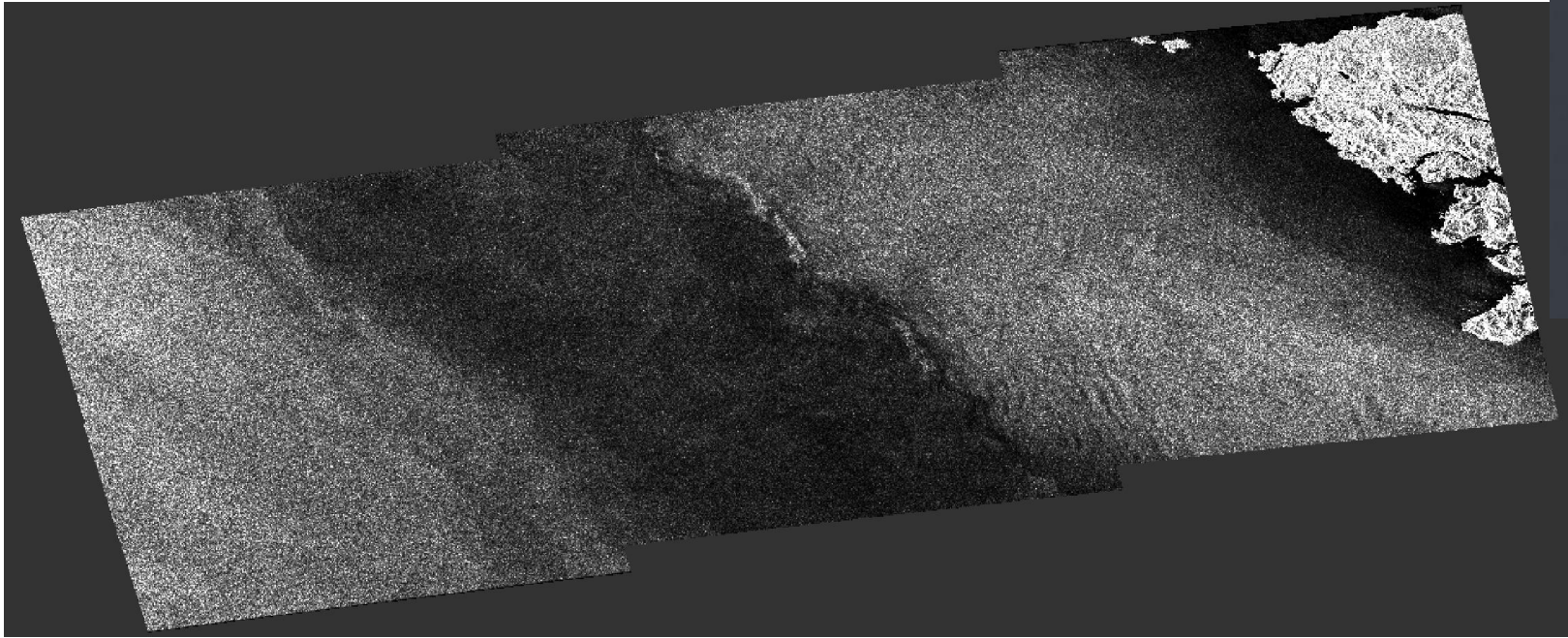
- Bright regions usually mean rough water -> wind or waves
- Dark regions usually mean smooth water -> low wind, surface film
- Complex signatures -> atmospheric imprints, convection, rain fronts
- Dynamic features -> fronts, jets eddies



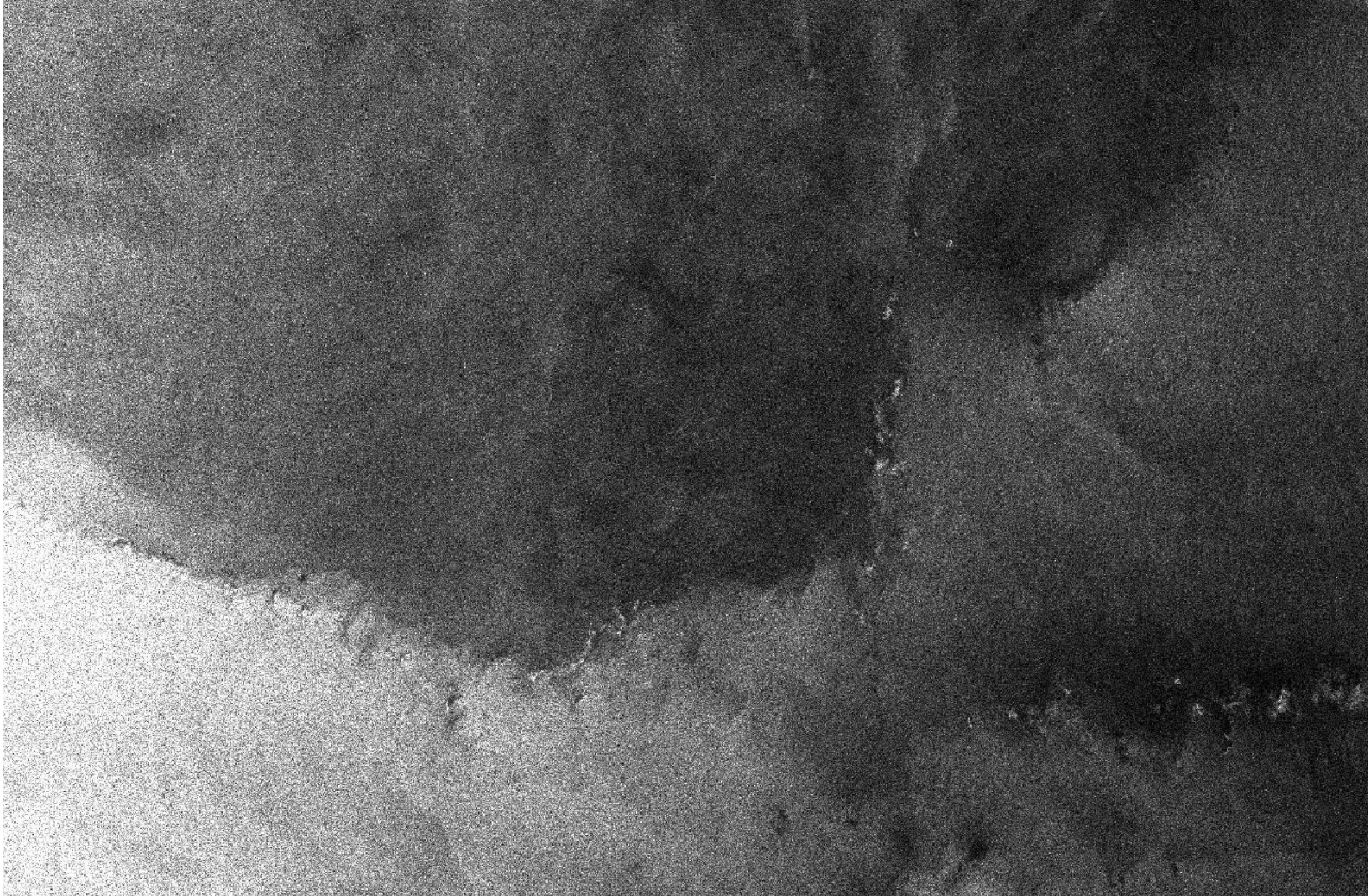
Some examples of SAR imagery

- For each image write down what you observe
- Hypothesize on what is leading to the variability of tone and texture you see in the image

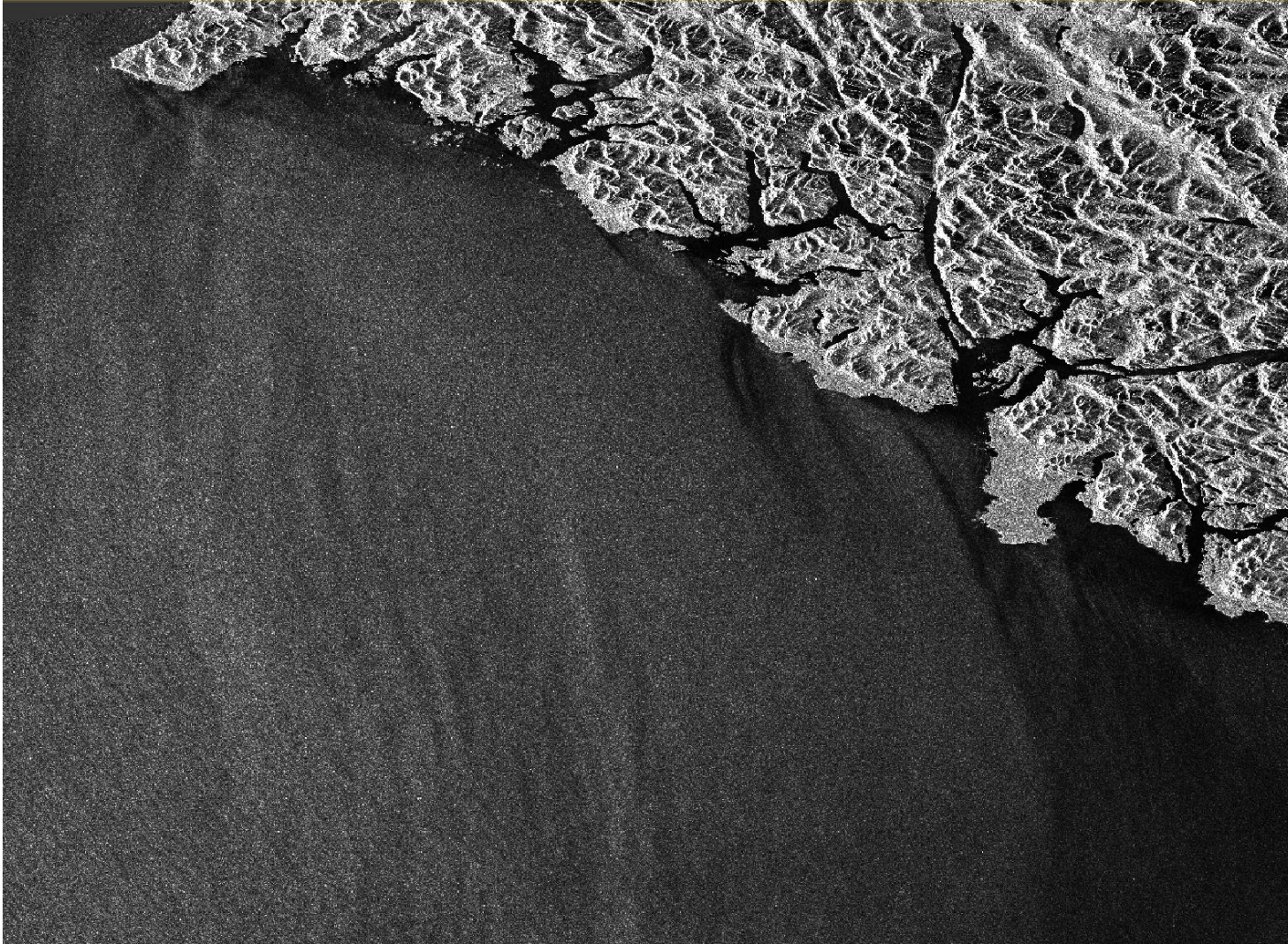
Example: May 11, 2025



Example: May 17, 2025



Example: May 17, 2025



A dataset of examples:

"A labelled ocean SAR imagery dataset of ten geophysical phenomena from Sentinel-1 wave mode", Wang et. al (2019)

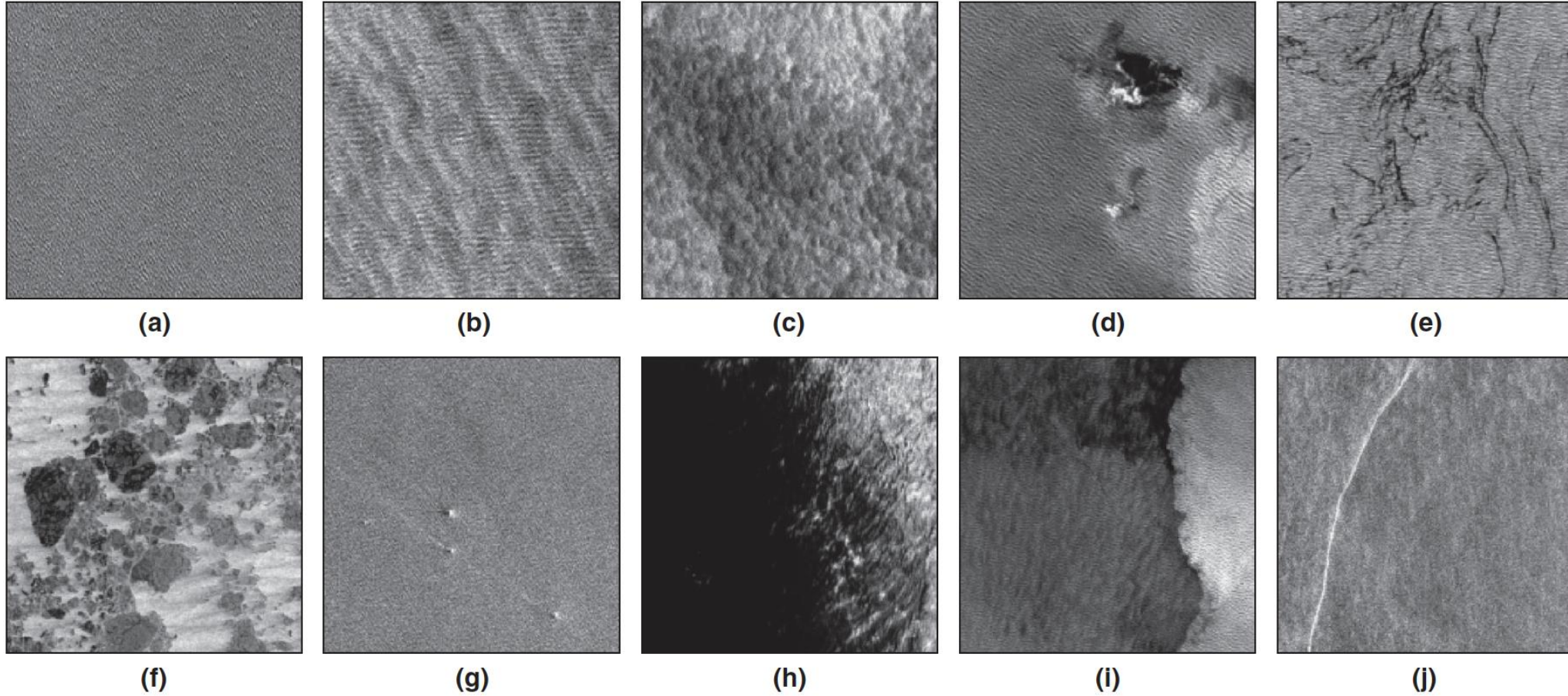


FIGURE 1 From (a) to (g) are image examples of pure ocean waves, wind streaks, micro convective cells, rain cells, biological slicks, sea ice, icebergs, low wind area, atmospheric front and oceanic front

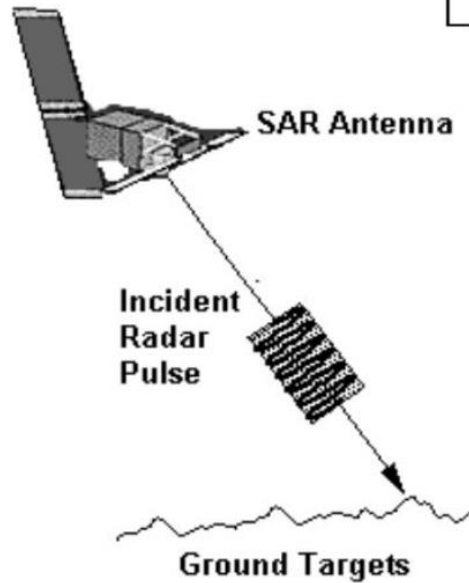
Fully Polarimetric SAR data

- These data represent the full information of the sent and received waves

Incident Wave

Jones vector of
incident wave (E^I)

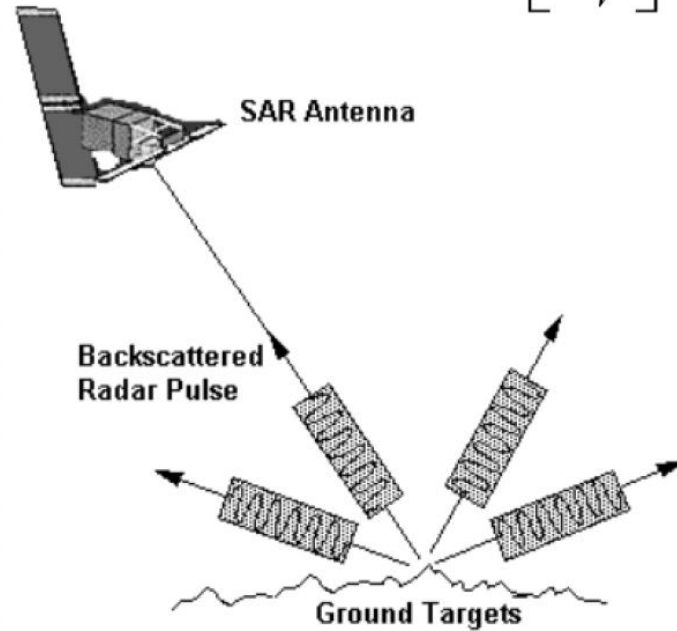
$$\begin{bmatrix} E_H^I \\ E_V^I \end{bmatrix}$$



Scattered Wave

Jones vector of
scattered wave (E^S)

$$\begin{bmatrix} E_H^S \\ E_V^S \end{bmatrix}$$



Recall:

$$E_H = |E_H| e^{j\phi_H}$$

$$E_V = |E_V| e^{j\phi_V}$$

The Scattering matrix

- the sent and received waves are related through the scattering matrix
- This matrix encodes the scattering physics -> for a material (sea ice and/or snow) that has both volume and surface scattering, this is rich information
- For the ocean, there are fewer applications

$$\begin{bmatrix} E_H^S \\ E_V^S \end{bmatrix} = \frac{e^{ikr}}{r} \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} E_H^I \\ E_V^I \end{bmatrix}$$

r = distance from target to receiver

k = wave number

λ = wavelength

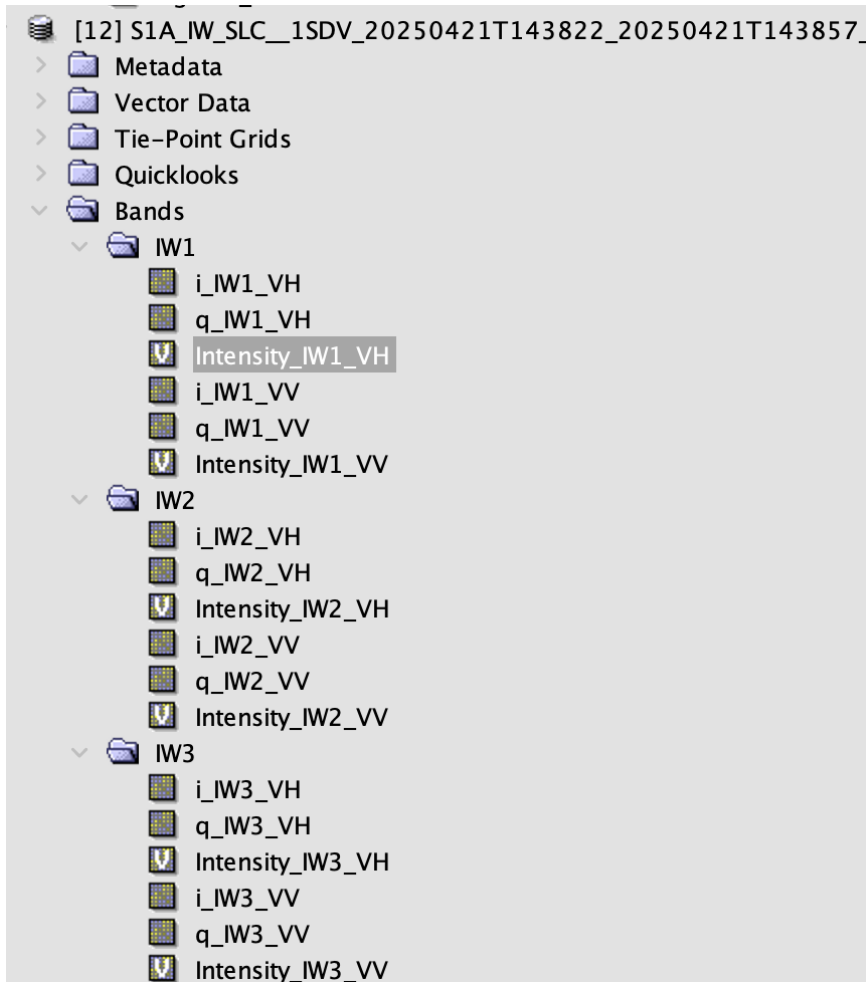
S = scattered

I = incident


$$k = \frac{2\pi}{\lambda}$$

Opening a fully polarimetric file

- These files are usually called SLC – single look complex
- They are large files – smaller swath and higher spatial resolution than dual pol



This is what you will see in the data file!

I is real part

Q is imaginary part

Processing these files is different than for dual-polarization

This will be in the second part of the SAR module

Further activities – I am happy to discuss and provide references

- Link the image features to physics or sensor characteristics
- Investigate SAR spectra to determine length scales of features
- Investigate wind speed retrieval algorithms over open water
- Investigate polarimetric decomposition
- Investigate other SAR sensors (e.g., Radarsat Constellation mission and SWOT) and other frequencies

References:

Wang, C., Mouche, A., Tandeo, P., Stopa, J. E., Long, P., N., Erhard, G., Foster, R., Vandemark, D. and Chapron, B. Labelled SAR imagery dataset of ten geophysical phenomena from Sentinel-1 wave mode Publisher: Sea scientific doi.org/10.17882/ 56796

Asiyabi, R.M., Ghorbanian, A., Tameh, S.N., Amani, M. Jin, S. and A. Mohammadzadeh, "Synthetic Aperture Radar (SAR) for Ocean: A Review," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 16, pp. 9106-9138, 2023, doi: 10.1109/JSTARS.2023.3310363

Rees, W.G., 2012, Physical Principles of Remote Sensing, Cambridge University Press.

Small, D, 2011, Flattening Gamma: Radiometric terrain correction for SAR imagery, *IEEE Transactions on Geoscience and Remote Sensing*, 49(8), 3081-3093

Van Zyl, Jakob and Yunjin Kim, Synthetic Aperture Radar Polarimetry, John Wiley & Sons, Inc., 2011